Real-Time Notification System

1.Introduction / Background

Real-time notification systems are widely used in modern web and mobile applications to instantly deliver updates such as messages, alerts, reminders, and activity notifications to users. These systems improve user engagement because users receive important information immediately without refreshing the application.

Many cloud platforms provide notification services such as **Firebase, OneSignal, PubNub, and Pusher**. These platforms provide infrastructure that allows developers to send notifications quickly and reliably across web and mobile applications.

2. Objective of the Project

The main objective of this project is to design a system that can send notifications to users instantly when an event occurs in the application.

The system aims to:

- Deliver real-time alerts to users.
- Improve user engagement and communication.
- Demonstrate how notification services work in web applications.

3. Technologies Used

List the technologies used in the project such as:

- Frontend: HTML, CSS, JavaScript, Bootstrap
- Backend: Java, Spring Boot, WebSocket, REST API
- Database: PostgreSQL
- Other Tools: Apache Maven, Git / GitHub, Postman.

4. System Description

The system works by detecting events in the application and sending notifications to

users instantly. First the user device registers with the notification service and receives a unique device token. When an event occurs, the backend server sends a notification request to the notification provider using that token.

The notification service processes the request and delivers the message instantly to the user's device or browser. This process ensures that users receive updates in real time.

5. Key Features / Functionalities

- User Registration and Login
- Device Registration for Notifications
- Real-Time Notification Delivery
- Event Detection System
- Notification API Integration
- Admin Monitoring

6. Expected Outcome

The expected outcome of this project is a system that can send notifications instantly to users whenever an event occurs. This system will improve communication between the application and the users by ensuring that updates are delivered quickly and efficiently.

7. Conclusion

Real-time notification systems play an important role in modern applications by providing instant communication between the system and users. This project demonstrates how notification platforms can be integrated into web applications to deliver updates efficiently and improve the overall user experience.